

UI-ATI REPORT

1. The AI-powered drone routing problem was solved using the A* Search algorithm. The algorithm combines the actual travel cost and heuristic estimate to identify the safest and shortest route while avoiding blocked paths. Compared to Greedy Best-First Search, A* provides better optimality, reliability, and adaptability in dynamic flood-affected environments, making it highly suitable for real-time emergency medicine delivery.
2. The smart city traffic optimization problem was solved using a Genetic Algorithm, which efficiently searches a large solution space by simulating natural selection. The algorithm improves traffic signal timings through selection, crossover, and mutation, reducing vehicle waiting time, fuel consumption, and congestion. Compared to Hill Climbing, the Genetic Algorithm avoids local optima, scales well to hundreds of intersections, and adapts effectively to dynamic traffic conditions, making it a suitable choice for real-time traffic management.
3. The Mars rover exploration problem was solved using an Online Search Agent, which enables the rover to make decisions while exploring an unknown environment. The rover continuously updates its internal map using sensor data, avoids hazards such as rocks and craters, and safely navigates toward the sample collection point. Compared with uninformed and informed search techniques, online search provides greater adaptability, handles partial observability effectively, and is well suited for real-time autonomous exploration under uncertain conditions.
4. The university exam timetable generation problem was modeled as a Constraint Satisfaction Problem (CSP) and solved using Backtracking Search with Forward Checking. The algorithm assigns time slots to subjects while satisfying constraints such as avoiding student exam overlaps, respecting faculty availability, and managing limited exam halls. Forward Checking reduces the search space by eliminating invalid choices early, making the scheduling process faster and more efficient. This approach is scalable, reliable, and well suited for generating valid exam timetables in large universities.
5. The real-time strategy game decision-making problem was solved using the Minimax algorithm with Alpha-Beta Pruning. Minimax evaluates possible game states by assuming optimal play from both the AI and the opponent, while Alpha-Beta Pruning improves efficiency by eliminating branches that cannot affect the final decision. An evaluation function based on resources, army strength, territory control, enemy

strength, and unit losses guides the AI in selecting the best move. This approach provides a good balance between optimality and real-time performance, making it highly suitable for complex strategy games.